

KIDUS WORKNEH

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SUMMARY

PhD student in Electrical and Computer Engineering, doing research in heterogeneous memory architecture, LLVM and Clang based compiler solutions, and hardware/software co-design for secure memory.

EDUCATION

University of Colorado Boulder

PhD, Electrical, Computer & Energy Engineering

Expected 2027

MSc, Electrical, Computer & Energy Engineering (GPA:3.8/4.0)

2022–2023

Addis Ababa University

BSc, Electrical & Computer Engineering (GPA:3.23/4.0)

2016–2021

SKILLS

C++

LLVM/Clang

Python

gem5

Git

L^AT_EX

WORK EXPERIENCE

Graduate Research Assistant

Jan 2022 – Present

University of Colorado Boulder, ECEE, Boulder, CO

- Designed NUMA-aware type extensions in C++ that allows programmers to have a finer grain control over thread and data allocation by using linux APIs and designing a custom memory allocator.
- Modified template specialization mechanisms in the Clang frontend to allow better thread and data locality for programs running on multi socket machines. This led to the submission of a paper to a top tier conference, OOPSLA 2025.
- Currently working on extending C++ types to describe secure memory placement by using pointer alias analysis in LLVM to efficiently reduce the trusted compute base of C++ programs.
- Developed a *library extraction* tool to improve re-usability and achieve code size reduction by collapsing functions of the same effect into one function that gets imported as a library for large code bases written in Python. This led to a paper publication in a programming language workshop, HATRA 2024.
- Built a hardware/software co-design solution using the gem5 simulator, that achieves fast crash recovery without sacrificing integrity protection of secure non volatile main memory devices. This led to a paper publication to a top tier conference, ASPLOS 2024.
- Worked on reducing the spatial overhead of maintaining encryption metadata using a memoization table on the CPU chip. This led to a computer architecture journal, CAL January 2024.
- Served as a teaching assistant for Computer Organization (ECEN 3593) in the fall semester of 2022.

Researcher

Jun 2025 – Aug 2025

University of Colorado Boulder (Physics) & Cerfe Labs, Boulder, CO

- Optimized current-voltage characterization workflows for CeRAM (correlated-electron RRAM) the Keithley instrumentation device.
- Refactored a C code base to apply a burst of pulse signals to characterize the memory window capability of the CeRAM device under stress.

Research Intern

May 2020 – Nov 2020

University of Michigan, Ann Arbor, MI

- Co-developed a privacy-enhanced computer architecture using gem5 that sequesters plaintext in a small structure in the CPU pipeline. This enables computing on sensitive data while encrypting results and input operands in all other micro-architectural structures.
- Organized a benchmark suite for evaluating different privacy-enhanced frameworks with varying computational complexity.
- Both works led to a paper publication at the hardware security conference, SEED 2022 & 2021 respectively.

PUBLICATIONS

- Workneh**, Kasprzykowski, Izraelevitz “A NUMA-aware Type Extension with Introspective Typing” OOPSLA (in progress), 2026.
- Bellur, Alghamdi, **Workneh**, Izraelevitz “Leroy: Library learning for imperative programming languages,” HATRA, 2024.
- Thomas, **Workneh**, McCarty, Izraelevitz, Lehman, Bahar “A midsummer night’s tree: Efficient and high performance secure SCM” *ASPLOS*, 2024.
- Thomas, **Workneh**, Ishimiwe, McKevitt, Curlin, Bahar, Izraelevitz, Lehman “Baobab Merkle Tree for Efficient Secure Memory” *IEEE Computer Architecture Letters*, 2024.
- Biernacki, Demissie, **Workneh**, Andargie, Austin “Sequestered Encryption: A hardware technique for comprehensive data privacy” *IEEE SEED*, 2022.
- Biernacki, Demissie, **Workneh**, Namomsa, Gebremedhin, Andargie, Reagan, Austin “VIP-Bench: A benchmark suite for evaluating privacy-enhanced computation frameworks” *IEEE SEED*, 2021.